Formula & concept guide

Page 1: choose the relationship, keep the units, and track direction.

Start every calculation
1. List what you know. 2. Identify what you need.
3. Choose a formula. 4. Substitute numbers with units.
5. Calculate. 6. Check the size, unit, and direction of the answer.

Find	Relationship	Units / what it means
Displacement	Δx = final position - initial position	Signed change in position (m). Distance is the whole path length.
Average speed	speed = total distance ÷ total time	m/s; direction is not included. Include all travel and stopping time.
Average velocity	vavg = displacement ÷ total time	m/s with a direction or a + / - sign.
Acceleration	a = (vf - vi) ÷ t	m/s²; vf = final velocity, vi = initial velocity; t is elapsed time.
Final velocity	vf = vi + a × t	m/s; use a constant acceleration during the stated interval.
Net force	Fnet = sum of signed forces	N. Choose a positive direction before adding forces on ONE object.
Force / acceleration	Fnet = m × a a = Fnet ÷ m m = Fnet ÷ a	N; kg; m/s². Use NET force, not automatically the applied force.
Weight	weight = m × g	N. In this packet, g ≈ 10 m/s² on Earth. Mass is in kg.
Momentum	p = m × v	kg·m/s, with direction. p is momentum; v is velocity.

Useful rearrangements: distance = speed × time; time = distance ÷ speed. For momentum: v = p ÷ m and m = p ÷ v.

Use changes, not clock readings
Elapsed time = final time - initial time. From 4 s to 11 s, the elapsed time is 11 - 4 = **7 s**. For a straight segment on a position-time graph: velocity = change in position ÷ change in time. Speed is the size of that result.

Explain it & solve it

Page 2: use the model that matches the situation.

Core idea	What to remember
Velocity & acceleration	Velocity is speed with direction. Acceleration is a change in speed OR direction. Turning at constant speed still means accelerating.
Net force & inertia	Balanced forces give zero acceleration: rest OR constant velocity. Inertia resists changes in motion; more mass means more inertia.
Newton’s three laws	1st: zero net force means constant velocity. 2nd: net force = mass × acceleration. 3rd: two interacting objects exert equal, opposite forces on EACH OTHER.
Gravity & friction	Mass measures amount of matter (kg); weight is gravitational force (N). Friction opposes sliding (or its tendency) between surfaces. Air resistance opposes motion through air.

Worked example 1 • Slowing down

A cyclist travels east at 7 m/s, then slows to 3 m/s in 2 s. Take east as positive.
1. Change in velocity: 3 - 7 = -4 m/s.
2. Divide by time: a = -4 ÷ 2 = **-2 m/s²** (west).
3. Check: acceleration is opposite the eastward velocity, so speed decreases.
A negative acceleration does not always mean slowing down. If velocity is also negative, negative acceleration makes the speed increase.

Worked example 2 • Opposing forces

A 12 kg cart is pushed right with 18 N; friction is 6 N left.
1. Choose right as positive. **2.** Fnet = 18 + (-6) = 12 N.
3. a = 12 ÷ 12 = **1 m/s² right**.
4. Check: the cart accelerates toward the greater force.

Momentum challenge • Two carts stick together

If outside forces have negligible effect during the collision:
1. Add initial momenta: $p_{total} = m_1v_1 + m_2v_2$ (keep signs).
2. Add masses: $m_{total} = m_1 + m_2$.
3. Shared final velocity = $p_{total} \div m_{total}$.
Example: 1 kg at 3 m/s meets 2 kg at rest. $v_{final} = (1 \times 3 + 2 \times 0) \div (1 + 2)$ = **1 m/s** in the original direction.

Read a graph
Distance-time: slope = speed; steeper means faster; flat means stopped.
Speed-time: change in speed ÷ time gives acceleration along a fixed direction. Flat means steady speed; the object may still turn.
Speed graphs do not show direction; signed position and velocity graphs can.